Type: MCQ

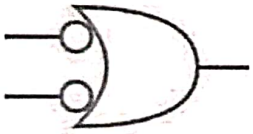
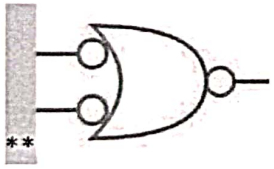
Q1. The Parity of the binary number 11011001 is (0.5)

1. Even
2. **Odd
3. Not known
4. Same as number of 0s

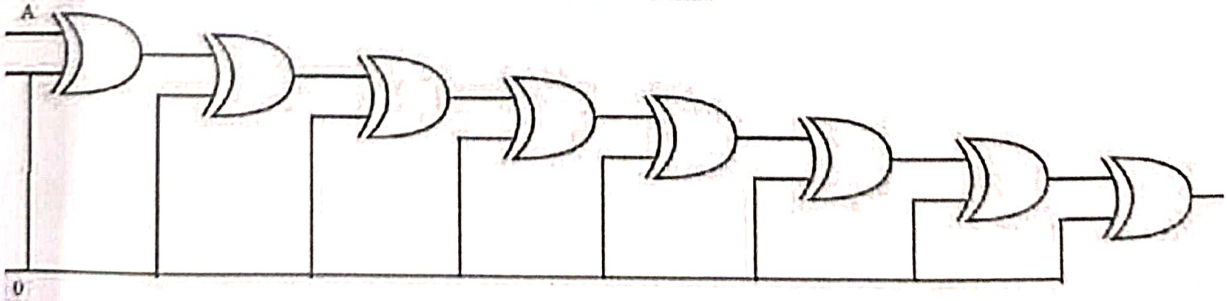
Q2. 2421 code is (0.5)

1. **Weighted self-complementing code
2. Non-weighted self-complementing code
3. Weighted non-self-complementing code
4. Non-weighted non-self-complementing code
5.

Q3. Which of the gates shown in the figure is an AND gate (0.5)

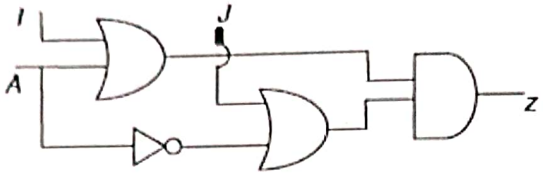
1. 
2. 
3. ** 
4. 

Q4. Find the output of the circuit shown in the figure (0.5)



1. 0
2. 1
3. **A
4. A'

Q5. The given circuit is used to implement the function z = f(A, B) = A' + B . What values are to be selected for I and J? (0.5)



1. I = 0, J = B
2. **I = 1, J =B
3. I = B, J = 1
4. I = B, J = 0

Q6. For a transitor in operation, the time taken for low to high transition is 0.2 mSec and the time taken for high to low transition is 0.1 mSec. The propagation delay of the (0.5)

1. 0.02 mSec
2. 0.3 mSec
3. 0.1mSec
4. **0.15 mSec

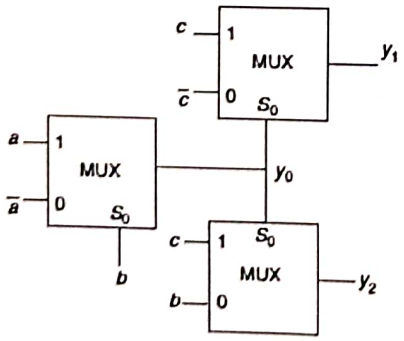
Q7. In a digital system, a high noise margin is desirable because it: (0.5)

1. Allows for faster data transmission
2. Reduces power consumption
3. **Increases the tolerance to voltage fluctuations and noise
4. Makes the circuit smaller and more compact

Q8. The carry generate and carry propagate function of the look-ahead carry adder is (0.5)

1. CG = A + B, CP = A $\oplus$ B
2. CG = A $\oplus$ B, CP = A + B
3. **CG = AB, CP = A $\oplus$ B
4. CG = AB, CP = A + B

Q9. The given circuit act as (0.5)



1. **full adder
2. half adder

3. full subtractor
4. half subtractor

Q10. Reflected binary code is also known as _____ (0.5)

1. BCD code
2. Binary code
3. ASCII code
4. **Gray Code

Type: DES

Q11. Implement a full subtractor using **only half subtractors**. (4)

Q12. Implement active high 4 to 16 decoder using **only two 2 to 4 active high decoders** and AND gates. (4)

Q13. Design a the following 2-input gates using Full Adder. (3)

i. XOR gate

ii. AND gate

iii. OR gate

Q14. Implement 2-input XNOR gate using minimum number of NAND gates. (3)

Q15. Design a 1-bit magnitude comparator with necessary logic gates. Use this to implement a 3-bit magnitude comparator. (3)

Q16. Implement the function $F(A,B,C,D) = \sum m(2,3,7,8,10)$ using 8 to 1 Multiplexer. (3)

Q17. Design a 3-bit binary to grey code converter with necessary logic gates. (3)

Q18. Obtain the reduced POS expressing using K-map for $\sum m(3,5,6,7)$. (2)